NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 07

CI/CD and Release

Verified pipeline behaviour, build and test gates, run history, environments and the release checklist.

Document control

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Intended readers	Developers and whoever performs releases; assessor checking delivery automation
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

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1 Pipeline inventory

Three GitHub Actions workflows live in `.github/workflows/`. Every workflow sets `permissions: contents: read` Implemented. No deployment (CD) job exists; artifacts are built and uploaded, but nothing is deployed Implemented.

Workflows

ID	Workflow	Triggers	Jobs and gates
CI-01	`android-ci.yml` "Android CI"	Push to `main`, `master`, `release/**`; PR to main/master; manual	**backend-test** (15 min): Node 22 (since v2.0; was 20), `npm ci`, `npm test` with a test-only JWT secret. **build** (30 min): JDK 17 Temurin, Gradle setup, wrapper validation, `ktlintCheck` (since v2.0), `./gradlew test`, `:app:assembleDebug`, upload `nexus-debug-apk` (14 days). Only if repository variable `NEXUS_API_URL` is set: `assembleRelease bundleRelease` and upload `nexus-release-artifacts`.
CI-02	`backend-ci.yml` "Backend CI"	Same branches, filtered to `backend/**` and its own file; manual	Node 22 tests, then `docker build`, run the container with `NODE_ENV=production`, wait 3 s, and require HTTP 200 from `/health`; the container is always removed.
CI-03	`blank.yml` "CI"	(removed in v2.0)	GitHub starter template that only echoed text; deleted (AUD-13).

Not gated in CI: detekt (AUD-23), Kover coverage thresholds, instrumented (`androidTest`) tests, and dependency vulnerability scanning Implemented. Dependabot opened PR #27, but the repository has no `.github/dependabot.yml`, so its configuration lives in GitHub settings Unknown.

2 Verified run history

Retrieved with `gh run list` / `gh run view` on 2026-11-05 (78 most recent runs: 49 success, 29 failure).

Notable runs

Date (UTC)	Trigger	Result	Cause (from run logs / annotations)
2026-09-22	Several pushes/PRs	Android CI mixed pass/fail	Early iterations; not analysed further
2026-09-24 07:17	Push to main (PR #24 merge)	**Android CI success**	Last successful Android CI on `main`
2026-09-26 18:23	PR #25 and its merge	All workflows failed in 3–5 s	Annotation: "The job was not started because your account is locked due to a billing issue."
2026-10-04 18:13	PR #26 and merge (baseline 3a3f275)	All workflows failed in 3–4 s	Same annotation; jobs never started. **The baseline code has never been verified by CI.**
2026-10-31 08:37	Dependabot PR #27 (zod 3.25.76 → 4.6.6)	Android build job passed; both backend test jobs **failed**	`TypeError: Cannot read properties of undefined (reading 'map')` at `server.js:133`. The Docker smoke test did not run. See AUD-03.

The Backend CI Docker smoke test (CI-02) has no successful run in the retrieved history Implemented. Its correctness is Inferred from reading the workflow only.

3 Local verification for this package

Commands run on 2026-11-05 on the author's Windows workstation

Command	Result
`cd backend; npm test` at v1.0 (zod 3.25.76)	2 suites, **16/16 passed**, 31.7 s
`cd backend; npm test` at v2.0 (zod 4.6.7)	3 suites, **29/29 passed**; `npm audit --omit=dev` : 0 vulnerabilities
`./gradlew :app:testDebugUnitTest --rerun` at v1.0 (JDK 21 runtime)	BUILD SUCCESSFUL; 7 classes, **32/32 passed**
`./gradlew :app:testDebugUnitTest` at v2.0	BUILD SUCCESSFUL; 10 classes, **42/42 passed**
`./gradlew :app:connectedDebugAndroidTest` at v2.0 (Android 7.0 emulator)	**16/16 passed** with the API stopped; 14/16 when a live API was reachable (AUD-22)
`assembleRelease -PNEXUS_API_URL=https://api.nexus.invalid/api/` at v2.0	APK signed with the upload key, verified by `apksigner` (APK Signature Scheme v2). A configuration-cache error in the HTTPS guard was fixed, after which the build is SUCCESSFUL; an http:// URL is rejected. The placeholder build was deleted.
`assembleDebug` at v2.0	Succeeded; installed and exercised on the emulator
Docker build	Not run Unknown. The Docker daemon was not running; `docker compose config` validates both compose files.

Details per suite are in NX-09-TDH.

4 Environments

Known environments

Environment	API URL	Status
Local emulator	`http://10.0.2.2:5263/api/` (debug default)	Implemented Documented in `docs/ONLINE_OFFLINE.md`
Local phone on LAN	`-PNEXUS_API_URL=http://<LAN-IP>:5263/api/`	Implemented Debug builds only
CI	None (tests use Supertest in process; smoke test uses localhost)	Implemented
Staging / production	None	Implemented Not provisioned; the README states the earlier production hostname "was not a configured service" (CON-02)

5 Release process

No release has been published: `gh release list` returns nothing and `versionCode` is 1 **Implemented**. The Gradle build enforces two release rules: `NEXUS_API_URL` must start with `https://` (`preReleaseBuild` check), and release builds minify and shrink resources **Implemented**. Since v2.0 a `release` signing config reads `keystore.properties` (git-ignored) or `NEXUS_KEYSTORE_*` environment variables, and the upload key was created with `scripts/create-upload-keystore.ps1` (AUD-08 fixed). CI does not have the key yet, so CI release artifacts remain unsigned until the secrets are added.

5.1 Release checklist

Items marked **Recommendation** are recommendations for the first release. The others restate rules the repository already enforces or documents.

1. CI (CI-01, CI-02) is green on the release commit; this requires IMP-01. **Recommendation**
2. Backend tests pass locally: `npm ci && npm test`.
3. Android unit tests pass: `./gradlew test`. Instrumented tests pass on at least one API 24 and one current device. **Recommendation**
4. Run the offline acceptance check in `docs/ONLINE_OFFLINE.md` (six steps) against the target API.
5. Increment `versionCode` / `versionName` in `app/build.gradle.kts` (2 / 2.0.0 for the final PoE). Settings shows `BuildConfig.VERSION_NAME`.
6. Hosted API is healthy over HTTPS with persistent storage and a stable `JWT_SECRET` (NX-11-DEP).
7. Set repository variable `NEXUS_API_URL` to the HTTPS URL ending in `/`.
8. Sign the APK/AAB with the release keystore held outside the repository (IMP-07). **Recommendation**
9. Do not seed test accounts on a public host (AUD-14). **Recommendation**
10. Add a dated entry to `PROJECT_IMPROVEMENTS.md`; tag the commit and create a GitHub release with the signed artifacts. **Recommendation**